Marking Scheme

2026 · Paper 1 · 105 marks

General Rules
- M / A / PROOF mark types
- **f.t.**：承上題錯誤但方法正確者，可獲後續步驟分。
- Deduction cap：A(1)=1 · A(2)=1 · B=0 (u, pp)
- u = 單位遺漏或錯誤；pp = 準確度不符要求。每部分最多扣 1 分。
- 答案正確且步驟合理者可獲滿分。
- 符號書寫略有出入但意思清楚者不扣分。 · Benefit of doubt
- 分數、小數或根式形式均接受，除非題目另有指明。

Section A(1) (35 marks)

1. Laws of Indices [B-]

a (3 marks)

M1 $(u^5v^{-3})^2 = u^{10}v^{-6}$

M1 $= u^{10-(-4)}\,v^{-6-5}$ f.t.

A1 $= \dfrac{u^{14}}{v^{11}}$ f.t.

Answer must carry positive indices; $u^{14}v^{-11}$ *alone scores 0A.*

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2. Factorization [B-]

a (1 mark)

A1 $4c^2 - 9d^2 = (2c+3d)(2c-3d)$

b (2 marks)

M1 $4c^2 - 9d^2 - 2c - 3d = (2c+3d)(2c-3d) - (2c+3d)$ f.t.

A1 $= (2c+3d)(2c-3d-1)$ f.t.

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3. Approximation [B]

a (1 mark)

A1 8000

b (1 mark)

A1 8264.058

c (1 mark)

A1 8264

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4. Algebraic Manipulation [B]

a (3 marks)

M1 $a = \dfrac{3b}{4} and c = \dfrac{3b}{2}$

M1 $\dfrac{2a+c}{b+c} = \dfrac{\frac{3b}{2}+\frac{3b}{2}}{b+\frac{3b}{2}}$ f.t.

A1 $= \dfrac{6}{5}$ f.t.

Accept 1.2 .

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5. Percentages [B]

a (4 marks)

- M1 $Let x be the number of paperback books sold.$
- M1 $x - (1 - 35\%)x = 224$
- A1 $x = 640$ f.t.
- A1 $Number of hardcover books sold = (1 - 35\%)(640) = 416$ f.t.
  Answering 640 (paperback) only scores 0 for this mark.

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6. Inequalities [B]

a (3 marks)

- M1 $5x + 12 \leq 2x gives 3x \leq -12$
- A1 $x \leq -4 or x > 8$
- A1 $The solution of (*) is x \leq -4 or x > 8$ f.t.

b (1 mark)

- A1 -4 f.t.

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7. Quadratic Equations [B+]

a (2 marks)

- M1 $\Delta = 0 : (-30)^2 - 4(9)(d) = 0$
- A1 $d = 25$

b (3 marks)

- M1 $9x^2 - 30x + 25 - 100 = 0$ f.t.
- M1 $3(3x + 5)(x - 5) = 0$ f.t.
- A1 $x = -\frac{5}{3} or x = 5$ f.t.
  Both intercepts required for the A mark.

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8. Angles and Parallel Lines [B+]

a (2 marks)

- M1 $\angle PRS = \angle SPR = 34°(base \angles, isos. \triangle) and \angle PRQ = \angle PTS = 47°(corr. \angles, ST \parallel QR)$
- A1 $\angle SRQ = 47° - 34° = 13°$ f.t.

b (3 marks)

- M1 $\angle UST = \angle SRQ = 13°(alt. \angles, ST \parallel QR)$ f.t.
- M1 $\angle SUT = 180° - 13° - \theta(\angle sum of \triangle)$ f.t.
- A1 $\angle QUR = \angle SUT = 167° - \theta(vert. opp. \angles)$ f.t.

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9. Statistical Measures [B+]

a (3 marks)

- A1 $Mean = 2.5$
- A1 $Median = 2.5$
- A1 $Standard deviation = 0.866$
  Accept $\sqrt{0.75}$ or $\frac{\sqrt{3}}{2}$.
  r.t. 3 sig. fig.

b (2 marks)

- M1 $New median = 3$ f.t.
- A1 $The median increases by 0.5.$ f.t.
  u-1 if the direction (increase) is omitted.

Section A(2) (35 marks)

10. Partial Variation [B]

a (4 marks)

M1 $W = A + Bh^2, where A and B are non-zero constants$

M1 $A + 16B = 55 and A + 81B = 250$

A1 $A = 7, B = 3 so W = 7 + 3h^2$ f.t.

A1 $W = 7 + 3(6)^2 = 115g$ f.t.

u-1 if the unit (g / grams) is omitted.

b (2 marks)

M1 $Weight of an ornament of height 8cm = 7 + 3(8)^2 = 199g; total weight of two ornaments of height 6cm = 2(115) = 230g$ f.t.

A1 $Since 199 < 230, the claim is NOT correct.$ f.t.

A mark is withheld unless a numerical comparison is shown.

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11. Statistical Measures [B+]

a (4 marks)

M1 $Q_1 = 32 and Q_3 = 48$

A1 $Inter-quartile range = 48 - 32 = 16$ f.t.

M1 $Range = 3(16) = 48, so (60 + k) - 21 = 48$ f.t.

A1 $k = 9$ f.t.

b (2 marks)

M1 $Mode = 48; number of participants with time \geq 48 is 6$ f.t.

A1 $Required probability = \frac{6}{19}$ f.t.

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12. Mensuration [B+]

a (3 marks)

M1 $Volume of the whole cone = \frac{1}{3}\pi(9)^2(12) = 324\pi cm^3$

M1 $Volume of the upper cone of height 8cm = 324\pi \times \left(\frac{2}{3}\right)^3 = 96\pi cm^3$ f.t.

A1 $Volume of the bottom part = 324\pi - 96\pi = 228\pi cm^3$ f.t.

u-1 if cm^3 is omitted.

b (3 marks)

M1 $Slant height = \sqrt{9^2 + 12^2} = 15cm, so the curved surface area of the whole cone = \pi(9)(15) = 135\pi cm^2$

M1 $Curved surface area of the upper cone = 135\pi \times \left(\frac{2}{3}\right)^2 = 60\pi cm^2$ f.t.

A1 $Curved surface area of the bottom part = 135\pi - 60\pi = 75\pi cm^2$ f.t.

u-1 if cm^2 is omitted.

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13. Remainder Theorem [A-]

a (3 marks)

M1 $f(x) = (x^2 - 4)Q(x) + kx - 12 for some polynomial Q(x)$

M1 $f(-2) = 0 : -2k - 12 = 0$

A1 $k = -6$ f.t.

b (5 marks)

M1 $f(2) = (-6)(2) - 12 = -24$ f.t.

M1 $f(x) = a(x + 2)(x - 1)(x - r) for some constants a and r$

M1 $f(0) = 2ar = 24 and f(2) = 4a(2 - r) = -24$ f.t.

A1 $a = 3, r = 4, so f(x) = 3(x + 2)(x - 1)(x - 4)$ f.t.

A1 $The roots are -2, 1 and 4, which are all integers. The claim is correct.$ f.t.

The three roots must be stated explicitly before the conclusion is accepted.

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14. Equations of Circles and Locus [A]

a (3 marks)

M1 $x - coordinate of G = \frac{-6+18}{2} = 6, so G = (6,5)$

M1 $Radius = GA = \sqrt{(6+6)^2+5^2} = 13$ f.t.

A1 $(x-6)^2+(y-5)^2 = 169, i.e. x^2+y^2-12x-10y-108 = 0$ f.t.

b (6 marks)

A1 $(i)\Gamma is parallel to L (and \Gamma lies midway between L and \ell).$

M1 $(ii) Slope of L = \frac{5-0}{6-18} = -\frac{5}{12}$

A1 $(ii)\Gamma: \ y-0 = -\frac{5}{12}(x+6), i.e. 5x+12y+30 = 0$ f.t.

M1 $(iii) Distance from G to \Gamma = \frac{|5(6)+12(5)+30|}{\sqrt{5^2+12^2}} = \frac{120}{13}$ f.t.

M1 $(iii) Let M be the foot of perpendicular from G to AH; \sin\angle GAH = \frac{GM}{GA} = \frac{120/13}{13} = \frac{120}{169}$ f.t.

A1 $(iii)\angle GAH = 45.2^\circ < 50^\circ, so the claim is NOT agreed.$ f.t.

Accept 45.2° *or* 45.24° *; pp-1 if the angle is left as an inverse-sine expression only.*
r.t. 3 sig. fig.

Section B (35 marks)

15. Probability [B+]

a (3 marks)

M1 $Total number of ways = C_4^{17} = 2380$

M1 $Favourable ways = C_4^7 + C_4^8 = 35+70 = 105 (no selection of 4 black pieces is possible)$

A1 $Required probability = \frac{105}{2380} = \frac{3}{68}$ f.t.

Accept 0.0441 *(3 sig. fig.).*

b (2 marks)

M1 $Required probability = 1 - \frac{3}{68}$ f.t.

A1 $= \frac{65}{68}$ f.t.

Accept 0.956 *(3 sig. fig.).*

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16. Geometric Sequences [A-]

a (2 marks)

M1 $r^3 = \frac{1296}{48} = 27, so r = 3$

A1 $1st term = \frac{48}{3} = 16$ f.t.

b (3 marks)

M1 $S_n = \frac{16(3^n-1)}{3-1} = 8(3^n-1) > 5\times10^{12}$ f.t.

M1 $3^n > 6.25\times10^{11}, i.e. n > \frac{\log(6.25\times10^{11})}{\log 3} = 24.72\ldots$ f.t.

A1 $The least value of n is 25.$ f.t.

pp-1 if n *is not given as an integer.*

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17. Quadratic Functions [A]

a (2 marks)

M1 $h(x) = 3(x^2 + 4kx) + 15k^2 + 6 = 3(x + 2k)^2 - 12k^2 + 15k^2 + 6$

A1 $Vertex = (-2k,\ 3k^2 + 6)$ f.t.

b (4 marks)

M1 $D = (-2k + 3,\ 3k^2 + 6) and E = (-2k - 3,\ -3k^2 - 6)$ f.t.

M1 $If the circumcentre is (0, 2), then (3 - 2k)^2 + (3k^2 + 4)^2 = (2k + 3)^2 + (3k^2 + 8)^2$ f.t.

A1 $-24k - 24k^2 - 48 = 0, i.e. k^2 + k + 2 = 0$ f.t.

A1 $\Delta = 1^2 - 4(1)(2) = -7 < 0, so no real k satisfies the equation. Hence there is NO such point F.$ f.t.

The discriminant argument must be shown; a bare 'No' scores 0A.

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18. Circles and Tangents [A]

a (2 marks)

PROOF $\angle TPQ = \angle PRQ (\angle in alt. segment) and \angle PTQ = \angle RTP (common \angle)$

PROOF $\therefore\ \triangle PTQ \sim \triangle RTP (AA)$

2 marks: 1 for a correct pair of equal angles with reason, 1 for the AA conclusion.

b (5 marks)

M1 $(i) \frac{TP}{TR} = \frac{TQ}{TP}, so TR = \frac{1200^2}{900} = 1600 cm$

A1 $(i) QR = 1600 - 900 = 700 cm, circumference = 700\pi cm$ f.t.

u-1 if cm is omitted.

M1 $(ii) \frac{PQ}{RP} = \frac{TQ}{TP} = \frac{3}{4}; let PQ = 3t, RP = 4t$ f.t.

M1 $(ii) \angle QPR = 90^\circ (\angle in semi-circle), so (3t)^2 + (4t)^2 = 700^2 giving t = 140$ f.t.

A1 $(ii) Perimeter = 420 + 560 + 700 = 1680 cm = 16.8 m < 17 m, so the claim is NOT agreed.$ f.t.

u-1 if the unit conversion cm to m is not handled.

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19. Applications of Trigonometry [A]

a (2 marks)

M1 $\angle BAC = 180° - 42° - 63° = 75° and AC = \frac{45\sin 42°}{\sin 63°} = 33.79\ldots cm (sine formula)$

A1 $\angle CAD = 105° - 75° = 30°, CD = \sqrt{AC^2 + 28^2 - 2(AC)(28)\cos 30°} = 16.9 cm$ f.t.

u-1 if cm is omitted.

r.t. 3 sig. fig.

b (3 marks)

M1 $Area of \triangle ABC = \frac{1}{2}(45)(AC)\sin 75° = 734.5\ldots cm^2$ f.t.

M1 $Area of \triangle ACD = \frac{1}{2}(AC)(28)\sin 30° = 236.5\ldots cm^2$ f.t.

A1 $Area of the paper card = 971 cm^2$ f.t.

u-1 if cm^2 is omitted.

r.t. 3 sig. fig.

c (7 marks)

M1 $(i) Perpendicular distance from A to BC = 45\sin 42° = 30.11 cm$

M1 $(i) Shortest distance = 30.11 \times \sin 38°$ f.t.

A1 $(i) Shortest distance = 18.5 cm$ f.t.

u-1 if cm is omitted.

r.t. 3 sig. fig.

M1 $(ii)\cos\angle ACD = \frac{AC^2 + CD^2 - 28^2}{2(AC)(CD)}, so \angle ACD = 55.7°$ f.t.

M1 $(ii) Angle between CD and CB produced = 180° - (63° + 55.7°) = 61.3°$ f.t.

M1 $(ii)\sin\theta = \sin 38° \sin 61.3°, where \theta is the required angle$ f.t.

Accept the equivalent method: vertical height of $D \div CD$.

A1 $(ii)\theta = 32.7° > 30°, so the claim is NOT correct.$ f.t.

r.t. 3 sig. fig.